Software Quality and Project Management

Lab 5: Data Quality and Validation

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Task 1: Great Expectations

For this task, three expectations have been used. The first one is '**ExpectColumnValuesToBeBetween'.** This expectation is used on the 'education-num' column to ensure that all values fall within a reasonable range (0-20). This is because education levels are numeric and should not exceed logical bounds. The 2nd one is **ExpectColumnValuesToNotBeNull**. In the Google Colab file, this is used for the 'age' column. This is to ensure that there are no missing values, as incomplete data can negatively impact analysis and model performance. The last one is **ExpectColumnValuesToBeInSet.** Is used for the income column, which verifies that values belong only to the valid categories (<=50K and >50K). This ensures consistency in the target variable used for classification.

Task 2: CleanLab

The datapoint is probably mislabeled because its measurements look more like an entirely different type of flower than the one it was labeled as. The petal length and petal width given are quite large, which usually belong to one class; however, it was labeled as another. This mismatch was predicted as a different label by the model, and CleanLab flagged it as a possible error.

Task 3: CleanLab

1. Do these suspected anomalous data points match what you expect for their species? Why or why not?
    - No, they don't match because some feature values, for instance, petal length, are too large for their species, making them abnormal.

2. Which feature (sepal length, petal length, etc.) seems most unusual in these points?
    - Petel-lenth is the most unusual feature since it has values much higher than normal for some species.
3. How can you check if these values are truly anomalies using the original dataset?
    - You can compare with the original dataset using statistics or plots to see if these values fall outside the normal range.

https://github.com/Beba-N/SQPM-LAB-5.git